

Technical Documentation

BinRazali / CasaClick

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Overview

- **Mobile**: React Native app (BinRazali)
- **API**: Symfony (CasaClick) exposing `/api/*` and `/api/mobile/*`
- **DB**: MySQL via Doctrine
- **Realtime**: WebSocket push in local dev + polling fallback in production

Install (developer)

```
```powershell
cd BinRazali
npm install
```
```

Android:

```
```powershell
npm start
npm run android
```
```

Local backend (development)

- Start Symfony API via `npm run server` (expects local `websitedev` / `casaclick` folder setup per docs).
- Start WS server via `npm run ws:server` (or `npm run dev:all`).

Build (Android release)

```
```powershell
npm run android:release
```
```

Architecture (key modules)

- API config: `src/app/api/config.ts`
- API calls: `src/app/api/*`
- Auth state: `src/app/authSlice.ts` + `src/app/store.ts`
- Navigation: `src/navigations/*`
- Notifications UI: `src/screens/main/NotificationsScreen.tsx`
- Realtime WS client: `src/services/notificationWebSocket.ts`
- Local notifications: `src/services/localNotifications.ts`

Configuration

- Production API origin: `src/app/api/config.ts` `!`` `PRODUCTION_API_ORIGIN`
- Toggle prod vs local: `USE_PRODUCTION_API`
- USB/LAN connection modes: `ANDROID_CONNECT_MODE`

Backups

- **Database**: use MySQL dump from the server environment (Railway/local MySQL)
- **Uploads/media**: back up public uploads directory used by Symfony (if applicable)

Security

See `docs/SECURITY_NOTES.md`.

System Requirements — Software

This section is written for the **BinRazali (React Native)** mobile app + the **CasaClick Symphony API** stack.

Operating system

- **Development OS (tested)**: Windows 10/11
- **Mobile target OS**: Android

SDKs / runtimes

- **Node.js**: `\(>= 20\)` (see `package.json` !` engines.node`)`
- **npm**: comes with Node.js (use the version bundled with Node 20+)
- **React Native**: 0.83.1 (see `package.json`)`
- **React**: 19.2.0 (see `package.json`)`
- **Android SDK**: Android Studio SDK manager (needed to build/run Android)
- **JDK**: required for Gradle Android builds (Android Studio installs a compatible JDK; use that)

Key libraries (mobile)

- **Navigation**: `@react-navigation/native` , `@react-navigation/stack``
- **State**: `@reduxjs/toolkit` , `react-redux` , `redux-persist``
- **Google sign-in**: `@react-native-google-signin/google-signin``
- **Realtime**: WebSocket client (built-in RN `WebSocket`)` + WS dev server (`ws`)`
- **Local notifications**: `@notifee/react-native``

Backend stack (expected / documented)

- **Symphony**: serves `/api/` and `/api/mobile/`` routes (see `docs/API.md`)`
- **MySQL**: data store (see `docs/DEPLOYMENT.md`)`
- **JWT**: LexikJWTAuthenticationBundle pattern (see `docs/DEMO_SCRIPT.md`)`

Compatibility notes

- **Google sign-in**: requires correct OAuth client IDs (see `docs/GOOGLE_SIGNIN_ANDROID.md`)`
- **Local dev API**: uses `adb reverse`` for ports 8000 (API) and 8082 (WS) (see `docs/USB_REALTIME_SYNC.md`)`
- **Production**: set `USE_PRODUCTION_API = true`` in `src/app/api/config.ts`` to use Railway HTTPS API.

System Requirements — Hardware

Mobile device (Android)

- **Minimum**
 - **Android version / API**: Android 8.0+ (API 26+) recommended baseline for modern notification behavior
 - **RAM**: 3 GB
 - **Storage**: 300 MB free (APK + cache)
 - **Network**: Wi-Fi or mobile data (required)
- **Recommended**
 - **Android version / API**: Android 10+ (API 29+)
 - **RAM**: 6 GB+
 - **Storage**: 1 GB+ free (smoother caching, images)

Sensors / device features

- **Required**: Internet connectivity
- **Not required (unless you extend features)**: GPS, camera, NFC, fingerprint

Development machine

- **Minimum**
 - **CPU**: 4 cores

- **RAM**: 8 GB
- **Storage**: 10 GB free (Android SDK + build cache)
- **Recommended**
 - **CPU**: 8 cores
 - **RAM**: 16 GB+
 - **Storage**: SSD with 20 GB+ free

API level mapping (why it matters)

- **API 26+**: notification channels (Android requirement) !' used by local notifications.
- **API 28+**: improved background execution limits behavior.

Security notes (mobile + API)

Authentication

- **JWT bearer tokens** are used for protected routes.
- Mobile sends `Authorization: Bearer <token>` (see `docs/API.md`).
- **Google sign-in** exchanges a native `idToken` for a backend JWT.

Transport security

- **Production API** must be HTTPS (Railway origin is HTTPS in `src/app/api/config.ts`).
- Avoid using HTTP on public networks; use HTTP only for local dev with USB/LAN.

Secrets / environment variables

- Do not commit real secrets:
 - API keys (Paymongo, OAuth secrets)
 - JWT private keys
 - Database credentials
- Use `.env` / platform env vars (see `docs/DEPLOYMENT.md`).

Storage on device

- Tokens are persisted using `redux-persist` + `AsyncStorage`.
- Risk: `AsyncStorage` is not encrypted by default. For higher security, migrate token storage to encrypted storage (future hardening).

RBAC

- API must enforce roles (example: tenant-only endpoints returning **403** for staff/admin tokens; see `docs/API.md`).
- UI alone must not be relied on to block access.

Common app hardening checklist (demo-ready)

- `APP_ENV=prod`, `APP_DEBUG=0` on Symfony in production
- Rotate secrets if exposed
- Validate inputs server-side (email/password, payment references)
- Return safe error messages (no stack traces)

Performance notes (short)

Startup

- First launch is slower due to JS bundle + initial API probe.
- Subsequent launches benefit from persisted auth state (redux-persist).

Network

- API calls use an 8-second polling fallback for sync (`/api/mobile/sync/revision`).
- When WebSocket is enabled (local dev), notifications update instantly and reduces polling pressure.

UI

- Lists use `FlatList` (listings, notifications) to avoid rendering huge arrays at once.

Recommended measurements (for defense/demo)

- Time-to-login (tap login !' dashboard rendered): _____ ms
- Listings load time (open listings !' cards visible): _____ ms
- Notification delivery time:
 - WebSocket: ~1s (local)
 - Polling: "d 8s (production fallback)